Outputs

The following list contains only the successful and feasible cases. For each benchmark, we report the candidate dimension together with the corresponding output of Semi-Algorithm 3.

Fib

1. Candidate dimension 1:

$$V\big(x_1^2 + x_1x_2 - x_2^2 + 1\big) \cup V\big(x_1^2 + x_1x_2 - x_2^2 - 1\big)\,.$$

Example 13

1. Candidate dimension 1:

$$V(x_1) \cup \{(1,1),(2,2),(3,3)\}.$$

Example 21

1. Candidate dimension 1:

$$\begin{aligned} &V\big(x_4,\, x_2,\, x_1 + x_3 - x_5,\, x_3^2 + x_3x_5 - x_5^2 + 1\big)\\ &\cup V\big(x_5,\, x_3,\, x_1,\, x_2^2 + x_2x_4 - x_4^2 + 1\big)\\ &\cup V\big(x_4,\, x_2,\, x_1 + x_3 - x_5,\, x_3^2 + x_3x_5 - x_5^2 - 1\big)\\ &\cup V\big(x_5,\, x_3,\, x_1,\, x_2^2 + x_2x_4 - x_4^2 - 1\big)\,. \end{aligned}$$

2. Candidate dimension 2:

$$V(x_4,\, x_2,\, x_1 + x_3 - x_5) \cup V(x_5,\, x_3,\, x_1)\,.$$

3. Candidate dimension 4:

$$V(x_1 + x_3 - x_5)\,.$$

Fib1

1. Candidate dimension 2:

$$V\big(2x_1x_2x_3 - x_1^2 - x_2^2 - x_3^2 + 2\big)\,.$$

Fib2

1. Candidate dimension 2:

$$V\big(4x_1^2x_2 - 2x_1x_3 - x_2 + 76\big)\,.$$

Fib3

1. Candidate dimension 2:

$$V\big(x_1x_2x_3 - x_1^2 + x_1x_2 - x_2^2 + x_1x_3 + x_2x_3 - x_3^2 + x_1 + x_2 + x_3 + 15 \quad\big)\,.$$

Nagata

1. Candidate dimension 1:

$$V\big(x_3 - 5,\, x_2^2 + 5x_1 - 19\big)\,.$$

2. Candidate dimension 2:

$$V(x_3 - 5)\,.$$

Squares and Cube

1. Candidate dimension 1:

$$V\big(x_2^2 - x_1x_3,\, x_1x_2 - x_3,\, x_1^2 - x_2\big) \cup V\big(x_1 - x_2 - x_3,\, x_2^3 - x_3^2\big)\,.$$

Squares

1. Candidate dimension 1:

$$V(x_3 - 1,\, x_1 + x_2 + 2) \cup V(x_3,\, x_1 + x_2 + 1)\,.$$

2. Candidate dimension 2:

$$V(x_1 + x_2 + x_3 + 1)\,.$$

Yagzhev9

1. Candidate dimension 6:

$$V(x_7 - x_8 + 7,\, x_2 - x_4 + x_6,\, x_1 - x_3 + x_5)\,.$$

Yagzhev11

1. Candidate dimension 10:

$$V\big(x_3x_5x_7 - x_2x_6x_7 - x_3x_4x_8 + x_1x_6x_8 + x_2x_4x_9 - x_1x_5x_9 + 67 \quad\big)\,.$$

Example 4.1

1. Candidate dimension 2:

$$\begin{aligned} &V\big(x_5 + x_6,\, x_4,\, 2x_1 + x_2 + x_6, 4x_2^2 + 3x_2x_3 - 3x_3^2 - 3x_2x_6 + x_3x_6 - 2x_6^2 \quad\big) \\ &\cup \left\{ \left(0, \frac{1}{2}, \frac{7}{8}, -\frac{5}{16}, 0, 0\right), \left(\frac{1}{2}, \frac{5}{16}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{4}, -\frac{5}{8}\right), \left(-\frac{1}{4}, -\frac{5}{8}, \frac{1}{8}, -\frac{1}{2}, -\frac{5}{8}, -\frac{3}{8}\right) \right\}. \end{aligned}$$

Example 4.2

1. Candidate dimension 1:

$$\begin{aligned} &V(2x_5 - 9x_6 - 43,\, 2x_4 - 11x_6 - 51,\, x_3 - 8x_6 - 51,\, 2x_2 - 9x_6 - 57,\, 2x_1 - 7x_6 - 57) \\ &\cup V(5x_5 - 3x_6 - 63,\, 5x_4 - 8x_6 - 83,\, 5x_3 - x_6 - 121,\, 5x_2 - 3x_6 - 78,\, 5x_1 + 2x_6 - 43) \\ &\cup V(7x_5 - 12x_6 + 35,\, 7x_4 - 19x_6 + 70,\, 7x_3 - 17x_6 + 105,\, 7x_2 - 12x_6 + 91,\, 7x_1 - 5x_6 + 63) \\ &\cup V(2x_5 - 9x_6 - 25,\, 2x_4 - 11x_6 - 25,\, x_3 - 8x_6 - 27,\, 2x_2 - 9x_6 - 27,\, 2x_1 - 7x_6 - 35) \\ &\cup V(5x_5 - 3x_6 - 25,\, 5x_4 - 8x_6 - 35,\, 5x_3 - x_6 - 55,\, 5x_2 - 3x_6 - 50,\, 5x_1 + 2x_6 - 25) \\ &\cup V(7x_5 - 12x_6 + 11,\, 7x_4 - 19x_6 + 18,\, 7x_3 - 17x_6 + 15,\, 7x_2 - 12x_6 + 25,\, 7x_1 - 5x_6 + 25) \\ &\cup V(2x_5 - 9x_6 - 9,\, 2x_4 - 11x_6 - 13,\, x_3 - 8x_6 - 9,\, 2x_2 - 9x_6 - 7,\, 2x_1 - 7x_6 - 11) \\ &\cup V(5x_5 - 3x_6 - 19,\, 5x_4 - 8x_6 - 9,\, 5x_3 - x_6 - 13,\, 5x_2 - 3x_6 - 14,\, 5x_1 + 2x_6 - 9) \\ &\cup V(7x_5 - 12x_6 + 33,\, 7x_4 - 19x_6 + 54,\, 7x_3 - 17x_6 + 45,\, 7x_2 - 12x_6 + 33,\, 7x_1 - 5x_6 + 19) \\ &\cup V(2x_5 - 9x_6 - 27,\, 2x_4 - 11x_6 - 39,\, x_3 - 8x_6 - 33,\, 2x_2 - 9x_6 - 37,\, 2x_1 - 7x_6 - 33) \\ &\cup V(5x_5 - 3x_6 - 57,\, 5x_4 - 8x_6 - 57,\, 5x_3 - x_6 - 79,\, 5x_2 - 3x_6 - 42,\, 5x_1 + 2x_6 - 27) \\ &\cup V(7x_5 - 12x_6 + 57,\, 7x_4 - 19x_6 + 106,\, 7x_3 - 17x_6 + 135,\, 7x_2 - 12x_6 + 99,\, 7x_1 - 5x_6 + 57)\,. \end{aligned}$$

Example 4.3

1. Candidate dimension 3:
$$V(x_6,\, x_4,\, x_2) \cup V(x_5,\, x_3,\, x_1)\,.$$

Fib and Nagata

1. Candidate dimension 2:
$$V\big(x_3 - 5,\, x_4^2 + x_4x_5 - x_5^2 - 1,\, x_2^2 + 5x_1 - 19\big) \cup V\big(x_3 - 5,\, x_4^2 + x_4x_5 - x_5^2 + 1,\, x_2^2 + 5x_1 - 19\big)\,.$$
2. Candidate dimension 3:
$$V\big(x_3 - 5,\, x_2^2 + 5x_1 - 19\big)\,.$$
3. Candidate dimension 4:
$$V(x_3 - 5)\,.$$

Example 4.3 and Fib

1. Candidate dimension 4:
$$V\big(x_5,\, x_3,\, x_1,\, x_7^2 + x_7x_8 - x_8^2 - 1\big) \cup V\big(x_6,\, x_4,\, x_2,\, x_7^2 + x_7x_8 - x_8^2 + 1\big)\,.$$
2. Candidate dimension 5:
$$V(x_5,\, x_3,\, x_1) \cup V(x_6,\, x_4,\, x_2)\,.$$

Factorial

1. Candidate dimension 2:
$$V(x_1x_3 - x_1 - x_2)\,.$$

SquareExponential

1. Candidate dimension 2:
$$V\big(x_1x_3^2 - 2x_2\big) \cup \{(1,1,1)\}.$$

Example 4.3 and Factorial

1. Candidate dimension 5:
$$V(x_5,\, x_3,\, x_1,\, x_7x_9 - x_7 - x_8) \cup V(x_6,\, x_4,\, x_2,\, x_7x_9 - x_7 - x_8)\,.$$

Example 1.5.0.2

1. Candidate dimension 1:
$$V\big(x_1 - x_3,\, x_3^2 - x_2 + x_3\big)\,.$$
2. Candidate dimension 2:
$$V(x_1 - x_3)\,.$$

Example 1.5.0.3

1. Candidate dimension 2:
$$V(x_2 + 1)\,.$$

Example 1.5.0.4

1. Candidate dimension 1:
$$V(x_2 - x_3,\, x_1 - x_3)\,.$$